OpenGeoSys Training – KIGAM/ZJU 2026

Exercise 1: Pumping Well - Theis

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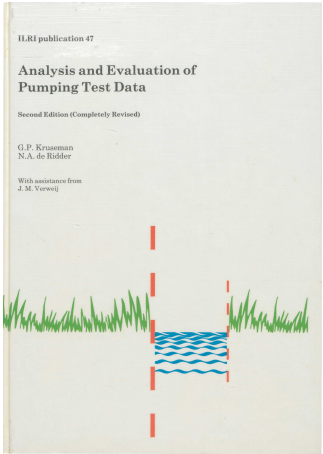
Daejeon/Hangzhou, 14th October 2026

DECOVALEX 2027

Agenda

1. OGS Course Overview
2. Theis solution
   - Analytical solution
   - Numerical solution: Axial symmetry
   - Numerical solution: 2D
   - Matrix leakage
3. References

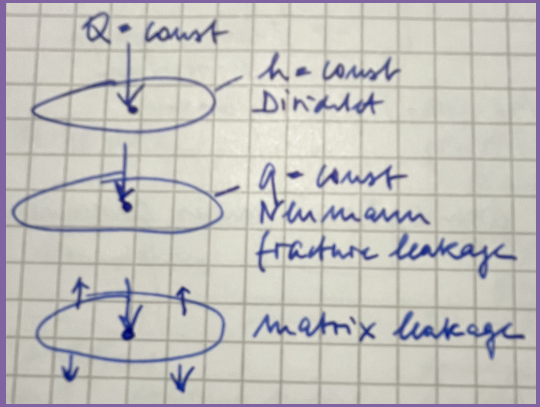
Book link: Kruseman and de Ridder, 1991



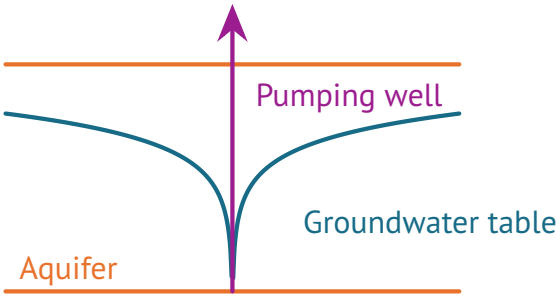
Exercise 1

Pumping wells

Theis solution



Theis problem examines the transient lowering of the water table induced by a pumping well. The assumptions required by the Theis solution are:



The aquifer:

- is homogeneous, isotropic, confined, and infinite in radial extent,
- has a uniform thickness and a horizontal piezometric surface.

The well:

- is fully penetrating the entire aquifer thickness,
- has a constant pumping rate,
- well storage effects can be neglected,
- no other wells or long-term changes in regional water levels.

The analytical solution of the drawdown as a function of time and distance:

$$s(r,t) = h_0 - h_r(r,t) = \frac{Q}{4\pi T} W(u) \quad \text{where} \quad u = \frac{r^2 S}{4Tt} \tag{1}$$

$$W(u) = -\gamma - \ln u + \sum_{k=1}^{\infty} \frac{(-1)^{k+1} u^k}{k \cdot k!} \approx -\gamma - \ln u \tag{2}$$

where

- s [L] is the drawdown or change in hydraulic head,
- h_0 [L] is the constant initial hydraulic head,
- h [L] is the hydraulic head at distance r at time t
- Q [L^3T^{-1}] is the constant pumping (discharge) rate,
- S [-] is the depth-integrated aquifer storage coefficient (volume of water released per unit decrease in h per unit area),
- T [L^2T^{-1}] is the transmissivity (how much water is transported horizontally per unit time).
- $\gamma = 0.577215664$ is the Euler-Mascheroni constant.

OGS project file: `axisym_theis.prj`

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<OpenGeoSysProject>
    <mesh axially_symmetric="true">axisym_theis.vtu</mesh>
    <geometry>axisym_theis.gml</geometry>
    <processes>
        ...
    <media>
        ...
    <time_loop>
        ...
    <parameters>
        ...
    <process_variables>
        ...
    <nonlinear_solvers>
        ...
    <linear_solvers>
        ...
</OpenGeoSysProject>
```

Listing: PRJ structure

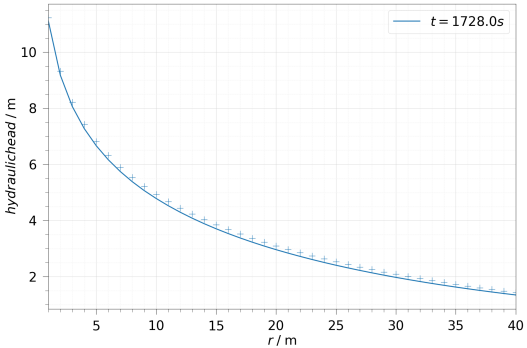
Theis solution #1

Exercise: Pumping wells: Comparison of analytical and numerical solutions

OGS benchmark gallery:

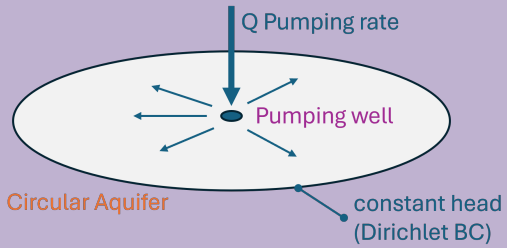
`https://www.opengeosys.org/6.5.9/docs/benchmarks/liquid-flow/axisym_theis/`

Course exercise on Github:



Repo:
`https://github.com/OlafKolditz/ogs-course-2026/tree/main/groundwater/well/theis1Dradial`

Theis with head boundary



Theis solution #2

Exercise: Pumping wells: Numerical solution: Meshing

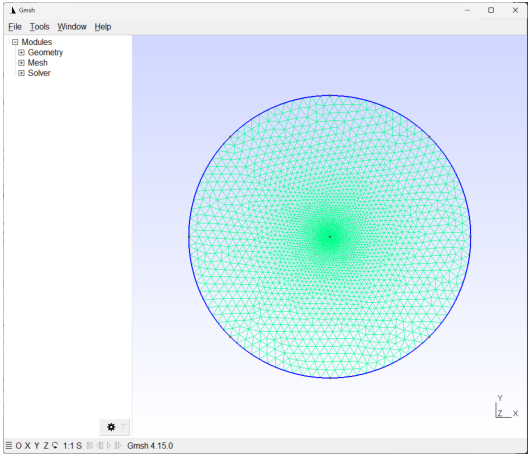


Fig.: Circle mesh: Triangulation with `gmsh`

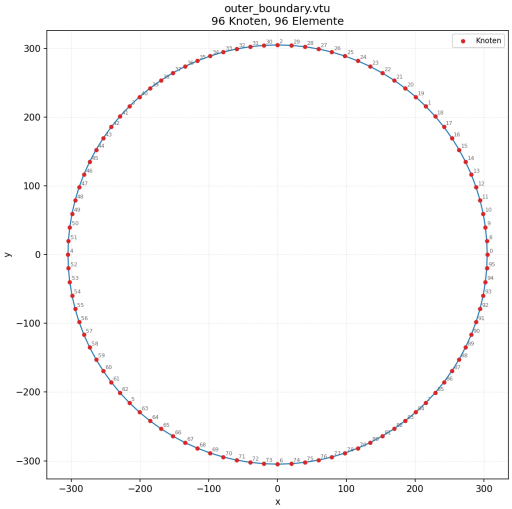
Procedure:

- meshing with with `gmsh` using various parameters (e.g. mesh density)
  > `meshing_gmsh.py`
- removing vertexes (elements of dimension 0)
  > `mesh_remove_vertexes.py`

Note: OGS can only use meshes without vertexes (elements of dimension 0)

Theis solution #2

Exercise: Pumping wells: Numerical solution: Boundary mesh



Procedure:

- extraction the boundary of the circle mesh (1)
  > `mesh_extract_boundary.py`
- the extracted boundary mesh is `outer_boundary.vtu`
- the boundary mesh can be visualized with (see figure left)
  > `plot_boundary_mesh.py`

The boundary extraction in necessary for assigning boundary conditions

OGS project file: `theis-2a-run.prj`

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<OpenGeoSysProject>
  <meshes>
    <mesh>circle_without_central_vertex_cell.vtu</mesh>
    <mesh>outer_boundary.vtu</mesh>
  </meshes>
    <geometry>theis-circle.gml</geometry>
    <search_length_algorithm>
        <type>fixed</type>
        <value>0.01</value>
    </search_length_algorithm>
    <processes>
        ...
    <media>
        ...
    <time_loop>
        ...
    <parameters>
        ...
</OpenGeoSysProject>
```

Listing: PRJ structure

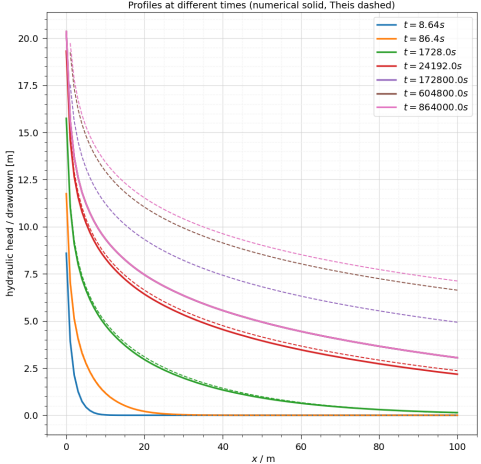
OGS project file:

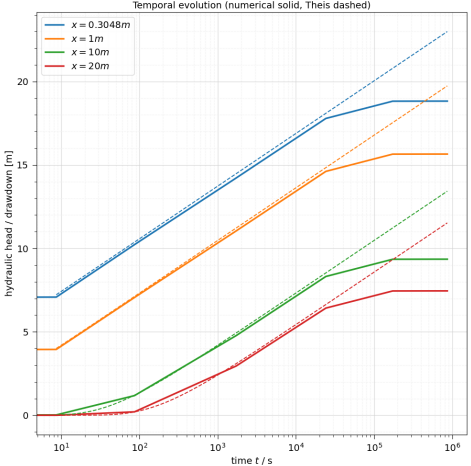
- why two meshes: domain and boundary
- set boundary conditions

Analysing results:
`plot_numerical_vs_analytical.py`

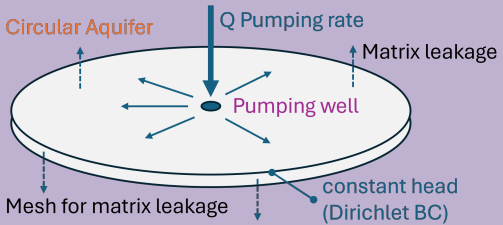
Theis solution #2

Exercise: Pumping wells: Numerical solution: OGS project





Matrix leakage



OGS project file: `ogs.prj`

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<OpenGeoSysProject>
  <meshes>
    <mesh>mesh_domain.vtu</mesh>
    <mesh>mesh_boundary.vtu</mesh>
    <mesh>mesh_leakage.vtu</mesh>
  </meshes>
    <geometry>well_injection.gml</geometry>
    <search_length_algorithm>
        <type>fixed</type>
        <value>0.01</value>
    </search_length_algorithm>
    <processes>
        ...
    <media>
        ...
    <time_loop>
        ...
    <parameters>
        ...
</OpenGeoSysProject>
```

Listing: PRJ structure

Meshes:

- model domain: triangulated circle
- model boundary: circle line elements
- leakage to matrix: triangulated circle for sink terms

Submeshes (e.g. for leakage) must contain references to the bulk mesh:
`> identifySubdomains`

Theis solution #3

Exercise: Matrix leakage

```xml
<?xml version="1.0" encoding="ISO-8859-1"?>
<OpenGeoSysProject>
  <meshes>
    <mesh>mesh_leakage.vtu</mesh>
  </meshes>
    <process_variables>
        <process_variable>
            <source_terms>
                <source_term>
                    <mesh>mesh_leakage</mesh>
                    <type>Volumetric</type>
                    <parameter>leakage_term</parameter>
                </source_term>
            </source_terms>
        </process_variable>
    </process_variables>
    <parameters>
        <parameter>
            <name>leakage_term</name>
            <type>Constant</type>
            <value>-1e-7</value>
        </parameter>
    </parameters>
</OpenGeoSysProject>
```

Listing: Leakage parameters

OGS project file: `ogs.prj`

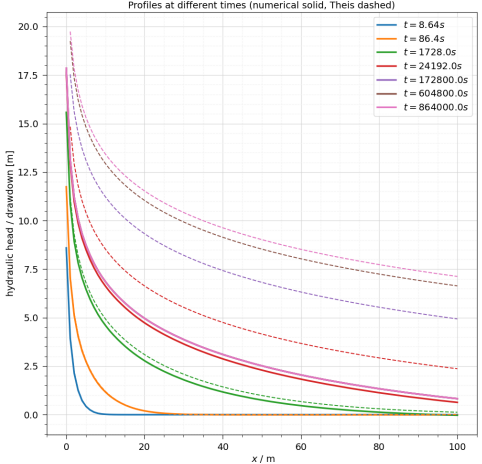
Leakage parameters:

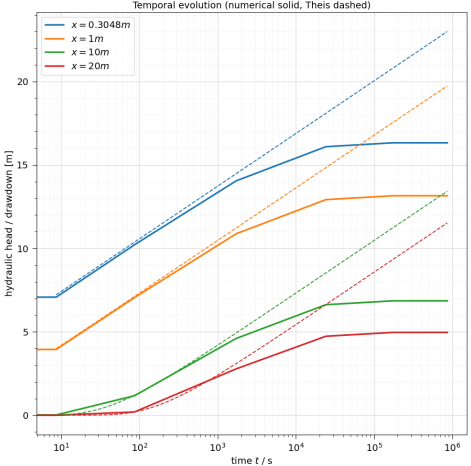
- Mesh link: `mesh_leakage`
- Parameter: `leakage_term`
- Type: `Volumetric`
- Value: `Constant -1e-7`

Repo link

Theis solution #3

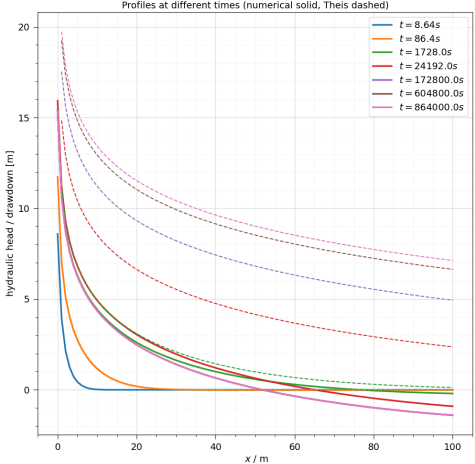
Exercise: Matrix leakage: Results: leakage term = -1e-7

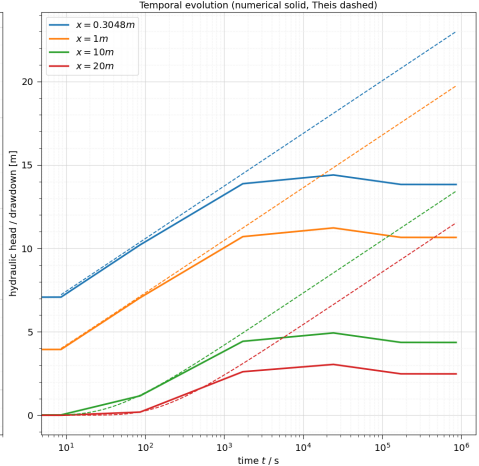




Theis solution #3

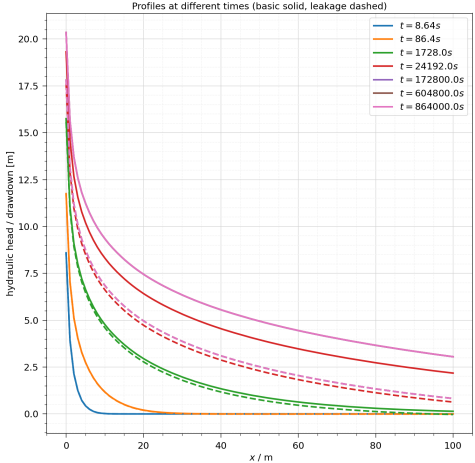
Exercise: Matrix leakage: Results: leakage term = -2e-7

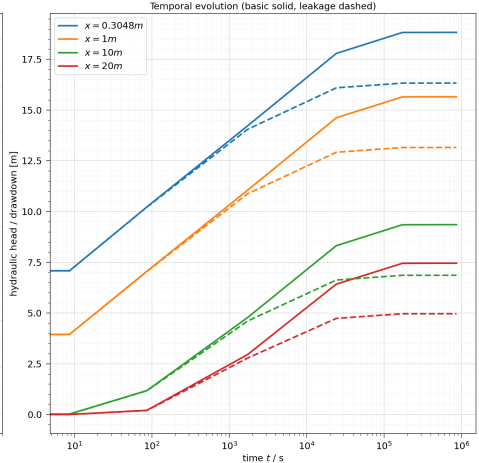




Theis solution #3

Exercise: Compare basic and leakage cases





Theis solution #3

Exercise: Time dependent matrix leakage

```xml
<?xml version="1.0" encoding="ISO-8859-1"?>
<OpenGeoSysProject>
  <meshes>
    <mesh>mesh_leakage.vtu</mesh>
  </meshes>
    <process_variables>
        <process_variable>
            <source_terms>
                <source_term>
                    <mesh>mesh_leakage</mesh>
                    <type>Volumetric</type>
                    <parameter>leakage_term</parameter>
                </source_term>
            </source_terms>
        </process_variable>
    </process_variables>
    <parameters>
        <parameter>
            <name>leakage_term</name>
            <type>Function</type>
            <expression>-1e-7/sqrt(t)</expression>
        </parameter>
    </parameters>
</OpenGeoSysProject>
```

Listing: Leakage parameters

OGS project file: `ogs.prj`

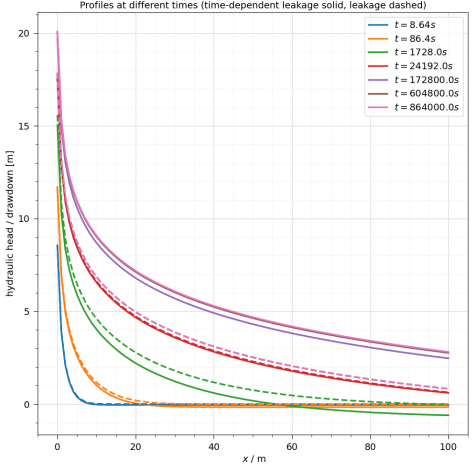
Leakage parameters:

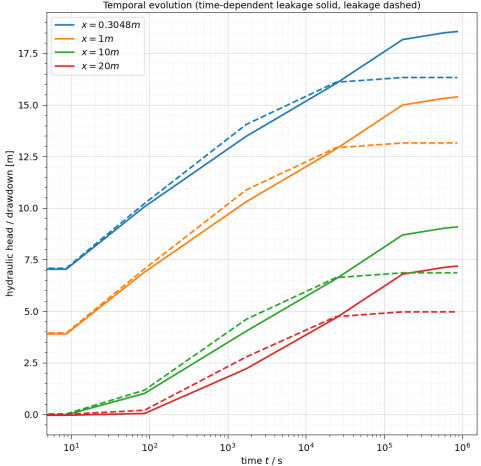
- Mesh link: `mesh_leakage`
- Parameter: `leakage_term`
- Type: `Volumetric`
- Function: `-1e-5/sqrt(t)`

Repo link

Theis solution #3

Exercise: Compare leakage cases: constant and time-dependent





Theis solution

ToDos

- using Python functions

References

[1] Olaf Kolditz. Computational Methods in Environmental Fluid Mechanics. Springer Berlin Heidelberg, 2002. ISBN: 9783662047613. DOI: 10.1007/978-3-662-04761-3. URL: http://dx.doi.org/10.1007/978-3-662-04761-3.

[2] G. P. Kruseman and N. A. de Ridder. 'Analysis and Evaluation of Pumping Test Data'. In: Soil Science Society of America Journal 35.4 (1971), pp. viii–viii. DOI: https://doi.org/10.2136/sssaj1971.03615995003500040014x. eprint: https://acsess.onlinelibrary.wiley.com/doi/pdf/10.2136/sssaj1971.03615995003500040014x. URL: https://acsess.onlinelibrary.wiley.com/doi/abs/10.2136/sssaj1971.03615995003500040014x.

[3] Erik Nixdorf, Mike Sips and Peter Morstein. 'Development of a digital workflow for layer reduction strategies and visual-analytical outcome analysis to enhance geo-hydraulic modelling efficiency'. In: Digital Water 3.1 (2025), pp. 1–24. DOI: 10.1080/28375807.2025.2460825.

[4] Charles V. Theis. 'The Relation Between the Lowering of the Piezometric Surface and the Rate and Duration of Discharge of a Well Using Ground-Water Storage'. In: Transactions, American Geophysical Union 16.2 (1935), pp. 519–524. DOI: 10.1029/TR016i002p00519.

[5] Zhi-Yuan Zhang, Christian Schmidt, Erik Nixdorf, Xingxing Kuang and Jan H. Fleckenstein. 'Effects of Heterogeneous Stream-Groundwater Exchange on the Source Composition of Stream Discharge and Solute Load'. In: Water Resources Research 57.8 (Aug. 2021), e2020WR029079. DOI: 10.1029/2020WR029079.

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